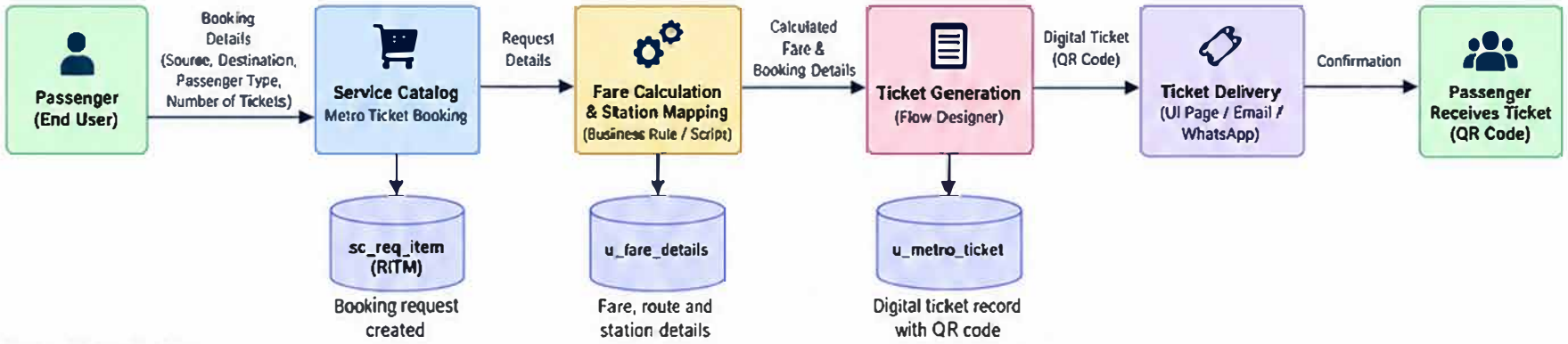
Project Design Phase-II
Data Flow Diagram & User Stories

Date	5 September 2026
Team ID	Individual
Name	Boyapati Rishi
Project Name	Digital Metro Ticket Booking System using ServiceNow
Maximum Marks	4 Marks

Data Flow Diagrams:

The Data Flow Diagram shows how a metro ticket booking request moves through the ServiceNow system, from passenger submission to ticket generation and delivery.

DFD Level 0 – Metro Ticket Booking Process



Data Flow Table:

Source / Actor	Process / System	Data Flow	ServiceNow Table / Record	Output / Next Step
Passenger (End User)	Service Catalog	Source, destination, passenger type, number of tickets	sc_req_item (RITM)	Booking request created
Service Catalog	Fare Calculation	Trip details	u_fare_details (custom table)	Calculated fare and route details
System (Business Rule)	Station Mapping	Source & destination	u_station_details (custom table)	Validates stations and calculates fare
Flow Designer	Ticket Generation	Approved booking details	u_metro_ticket (custom table)	Digital ticket with QR code
System	Ticket Delivery	Ticket details (QR code)	UI Page / Notification	Ticket sent to passenger
Passenger (End User)	Ticket Usage	QR code at metro station	Validation (at station entry)	Ticket verified and journey started

Data Flow Summary:

Passenger → Service Catalog → Fare Calculation & Station Mapping → Ticket Generation → Ticket Delivery → Passenger receives digital ticket (QR code).

User Stories

The following user stories define the functional requirements for the Digital Metro Ticket Booking System project.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Passenger (End User)	Service Catalog Booking	USN-1	As a passenger, I can book a metro ticket through the Service Catalog.	A request is created with source, destination, passenger type and number of tickets.	High	Sprint-1
Passenger (End User)	Input Details	USN-2	As a passenger, I can provide source, destination, passenger type and number of tickets in the catalog form.	Required fields are completed and the request can be submitted successfully.	High	Sprint-1
System / ServiceNow	Fare Calculation	USN-3	As the system, I can automatically calculate the fare based on source, destination and passenger type.	Fare is calculated correctly using configured fare logic.	High	Sprint-1
System / ServiceNow	Station Mapping	USN-4	As the system, I can validate source and destination stations and map the route details.	Only valid stations are allowed and route details are retrieved successfully.	High	Sprint-1
System / ServiceNow	Ticket Generation	USN-5	As the system, I can generate a digital ticket with a unique QR code after successful booking.	A ticket is created in the system with a unique QR code.	High	Sprint-1
Passenger (End User)	Ticket Access	USN-6	As a passenger, I can view and download my digital ticket with QR code.	The ticket is displayed with journey details and a valid QR code.	High	Sprint-2
Station Manager	Ticket Validation	USN-7	As a station manager, I can validate the QR code at the station entry.	The QR code is scanned and the ticket is verified as valid or invalid.	High	Sprint-2
Passenger (End User)	Booking History	USN-8	As a passenger, I can view my booking history.	All past bookings are listed with date, route and ticket status.	Medium	Sprint-2
Operations Team	Reporting & Analytics	USN-9	As a metro operations team member, I can view booking reports and analytics.	Reports show total bookings, revenue, peak usage times and popular routes.	Medium	Sprint-2
Passenger (End User)	Notifications	USN-10	As a passenger, I can receive notifications about my booking confirmation and ticket details.	Notifications are triggered via email/WhatsApp with booking and ticket information.	Medium	Sprint-2